

Question number	Answer	Mark
3(a)	A J	(1)

Question number	Answer	Additional guidance	Mark
3(b)	An explanation that includes the following points: • because this structure is very small (1) • and only an electron microscope has the ability to magnify this much (1)	Accept converse statement	(2)

Question number	Answer	Mark
3(c)	An explanation that includes the following points: • because the two membranes are very close together (1) • and the resolution of this microscope is not high enough (1)	(2)

Question number	Answer	Mark
3(d)(i)	An explanation that includes the following points: • mitosis results in two cells that each have a nucleus (1) • each bud contains a full set of chromosomes (1) • so the buds are genetically identical to the parent yeast cell (1)	(3)

Question number	Answer	Mark
3(d)(ii)	An answer that includes the following points: • because new cell structures need to be made in order for the bud to grow (1) • therefore oxygen is needed to produce the ATP (1) • therefore {glucose is needed to produce the ATP / amino acids are needed to make proteins} (1)	(3)